



Constellation

James A. FitzPatrick NPP
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Alexander Sterio
Site Vice President– JAF

JAFP-23-0066
December 1, 2023

United States Nuclear Regulatory Commission
Attn: Document Control Desk
Washington, D.C. 20555-0001

James A. FitzPatrick Nuclear Power Plant
Renewed Facility Operating License No. DPR-059
NRC Docket No. 50-333

Subject: LER: 2023-004, Misaligned Spacer Prevented Tie Breaker Closing Coil
Armature Operation on a Magne-Blast for EDG A Subsystem

Dear Sir or Madam:

This report is being submitted pursuant to 10 CFR 50.73(a)(2)(i)(B) and 10 CFR 50.73(a)(2)(v)(A).

There are no new regulatory commitments contained in this report.

Questions concerning this report may be addressed to Mr. Mark Hawes, Regulatory Assurance, at (315) 349-6659.

Sincerely,

Alexander Sterio
Site Vice President

ADS/MH

Enclosure: LER: 2023-004, Misaligned Spacer Prevented Tie Breaker Closing Coil
Armature Operation on a Magne-Blast for EDG A Subsystem

cc: USNRC, Region I Administrator
USNRC, Project Manager
USNRC, Resident Inspector
INPO Records Center (IRIS)

NRC FORM 366 (10-01-2023)		U.S. NUCLEAR REGULATORY COMMISSION			APPROVED BY OMB: NO. 3150-0104		EXPIRES: 03/31/2024							
LICENSEE EVENT REPORT (LER) (See Page 2 for required number of digits/characters for each block) (See NUREG-1022, R.3 for instruction and guidance for completing this form http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/) Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collections Branch (T-6 A10M), U. S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by email to Infocollections.Resource@nrc.gov, and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk Officer for the Nuclear Regulatory Commission, 725 17th Street NW, Washington, DC 20503; email: oira_submission@omb.eop.gov. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.														
1. Facility Name James A. FitzPatrick Nuclear Power Plant					☒ 050 ☐ 052		2. Docket Number 05000333		3. Page 1 OF 4					
4. Title Misaligned Spacer Prevented Tie Breaker Closing Coil Armature Operation on a Magne-Blast for EDG A Subsystem														
5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved					
Month	Day	Year	Year	Sequential Number	Revision No.	Month	Day	Year	Facility Name	☐ 050 Docket Number				
04	12	2023	2023 -	004 -	00	12	01	2023	N/A	N/A				
9. Operating Mode					10. Power Level									
1					100									
11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)														
10 CFR Part 20		☐ 20.2203(a)(2)(vi)		10 CFR Part 50		☐ 50.73(a)(2)(ii)(A)		☐ 50.73(a)(2)(viii)(A)		☐ 73.1200(a)				
☐ 20.2201(b)		☐ 20.2203(a)(3)(i)		☐ 50.36(c)(1)(i)(A)		☐ 50.73(a)(2)(ii)(B)		☐ 50.73(a)(2)(viii)(B)		☐ 73.1200(b)				
☐ 20.2201(d)		☐ 20.2203(a)(3)(ii)		☐ 50.36(c)(1)(ii)(A)		☐ 50.73(a)(2)(iii)		☐ 50.73(a)(2)(ix)(A)		☐ 73.1200(c)				
☐ 20.2203(a)(1)		☐ 20.2203(a)(4)		☐ 50.36(c)(2)		☐ 50.73(a)(2)(iv)(A)		☐ 50.73(a)(2)(x)		☐ 73.1200(d)				
☐ 20.2203(a)(2)(i)		10 CFR Part 21		☐ 50.46(a)(3)(ii)		☒ 50.73(a)(2)(v)(A)		10 CFR Part 73		☐ 73.1200(e)				
☐ 20.2203(a)(2)(ii)		☐ 21.2(c)		☐ 50.69(g)		☐ 50.73(a)(2)(v)(B)		☐ 73.77(a)(1)		☐ 73.1200(f)				
☐ 20.2203(a)(2)(iii)				☐ 50.73(a)(2)(i)(A)		☐ 50.73(a)(2)(v)(C)		☐ 73.77(a)(2)(i)		☐ 73.1200(g)				
☐ 20.2203(a)(2)(iv)				☒ 50.73(a)(2)(i)(B)		☐ 50.73(a)(2)(v)(D)		☐ 73.77(a)(2)(ii)		☐ 73.1200(h)				
☐ 20.2203(a)(2)(v)				☐ 50.73(a)(2)(i)(C)		☐ 50.73(a)(2)(vii)								
☐ OTHER (Specify in Abstract below or in NRC Form 366A).														
12. Licensee Contact for this LER														
Licensee Contact Mr. Mark Hawes, Regulatory Assurance									Telephone Number (Include Area Code) 315-349-6659					
13. Complete One Line for each Component Failure Described in this Report														
Cause	System	Component	Manufacturer	Reportable to IRIS	Cause	System	Component	Manufacturer	Reportable to IRIS					
D	EK	BKR	G080	N										
14. Supplemental Report Expected						15. Expected Submission Date			Month		Day		Year	
☒ No									☐ Yes (If yes, complete 15. Expected Submission date)					
16. Abstract (Limit to 1326 spaces, i.e., approximately 13 single-spaced typewritten lines)														
On April 12, 2023, with James FitzPatrick Nuclear Power Plant (JAF) was at 100% power the Emergency Diesel Generator (EDG) "A" and "C" tie breaker (71-10504) did not close during a performance of monthly full load surveillance test (ST-9BA). This was caused by binding of the closing coil armature of the breaker due to a misaligned spacer installed in June 2021. The cause of this condition is inadequate procedural guidance to ensure proper orientation/alignment of the close coil sleeve and spacers. Tie breaker failure to close rendered the "A" Emergency Diesel Generator (EDG) subsystem inoperable. It was determined that this deficiency began after the last successful test/actuation on March 13, 2023. This condition was prohibited by Technical Specifications (TS), which is reportable per 10 CFR 50.73(a)(2)(i)(B) and 10 CFR 50.73(a)(2)(v)(A). The condition was corrected when the 71-10504 Tie Breaker was replaced on April 13, 2023. A corrective action was completed to revise MP-054.01 and MP-054.03 to improve spacer installation process and ensure proper breaker actuation.														

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
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1. FACILITY NAME

James A. FitzPatrick Nuclear Power Plant

☒ 050☐ 052**2. DOCKET NUMBER**

05000333

3. LER NUMBER

YEAR

2023

SEQUENTIAL
NUMBER

- 004

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NARRATIVE**Background**

The Emergency Onsite Power Supply System [EISS identifier: EK] consists of two redundant Emergency Diesel Generator (EDG) subsystems. Each EDG subsystem consists of two EDG units. The capability of the two units in the subsystem to start, force parallel, and attain rated voltage and frequency together within 10 seconds, and run the necessary engineered safeguard loads, meets the requirements for the Emergency AC Power System.

The EDG logic system utilizes a force parallel design. This allows two EDG units to parallel onto the respective Emergency AC bus together to provide a combined load. The "A" subsystem tie breaker (71-10504) force parallels the "A" and "C" EDG units.

Event Description

On April 12, 2023, with James FitzPatrick Nuclear Power Plant (JAF) at 100% power the EDG "A" and "C" tie breaker (71-10504) did not close during a performance of monthly full load surveillance test (ST-9BA). The "B" EDG subsystem was verified to be operable. Troubleshooting identified that the issue was internal to the breaker. On April 13, the 71-10504 Magne-Blast circuit breaker was replaced and at 23:00 the "A" EDG subsystem was restored to service. Total out of service time was approximately 43 hours.

Event Analysis

The removed Magne-Blast circuit breaker was sent to General Electric (GE) for evaluation. The GE investigation identified that intermittent binding of the close coil armature occurred, which prevented the full stroke required to close the circuit breaker. The binding was repeated multiple times during the investigation. Visual inspection confirmed the closing coil appeared to be askew in the assembly and the spacers were not aligned correctly.

The misaligned spacer did not allow the closing coil sleeve to seat properly in the cutout of the support. This decreased the clearance required for the closing coil armature to move freely inside the full length of the assembly. The decreased clearance caused the armature to bind against the spacer and the inside of the sleeve. The armature does not experience binding in all positions of the armature. The armature was rotated and then it was identified that it only became bound in certain positions. A properly installed coil assembly will not become loose, and closing coil will be able to operate without binding.

It was determined that the misaligned spacer was installed in June 2021 when 71-10504 was replaced. All subsequent surveillance tests were successful from June 2021 until the failure on April 12, 2023. On October 19, 2023, JAF performed tests of the removed breaker closing coil armature to demonstrate how the armature rotates during each breaker actuation. It rotates at a regular fix quantity during each actuation; therefore, the specific position which caused binding was rotated to that position during the last test on March 13, 2023. Based on this evaluation, the "A" EDG subsystem tie breaker 71-10504 would not have closed between March 13 and April 12, a condition which would have caused the "A" EDG subsystem to be inoperable.

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When "A" EDG subsystem is inoperable, Technical Specifications (TS) 3.8.1, Condition B requires it be restored to operable status within 14 days. Condition F states that if the Completion Time of Condition B cannot be met, then the unit shall be in Mode 3 within 12 hours and Mode 4 within 36 hours. This condition was prohibited by TS, which is reportable in accordance with 10 CFR 50.73(a)(2)(i)(B).

Concurrent with this condition, as reported by LER-23-003, the 7-day fuel oil supply [DE] was not met between March 27, 2023, and April 25, 2023. In addition, ST-9BB testing of the "B" EDG subsystem on March 27, 2023, rendered the subsystem inoperable. TS 3.8.1 Condition E applies when two EDG subsystems are inoperable. When "A" and "B" subsystems were inoperable at the same time, between March 27 to April 13, this circumstance is reportable in accordance with 10 CFR 50.73(a)(2)(v)(A) as a condition which could have prevented the fulfillment of the EDG system safety function.

Cause

Emergency Diesel Generator "A" subsystem tie breaker 71-10504 failed to close due to binding of the closing coil armature due to a misaligned spacer. The cause of this event is inadequate procedural guidance, to ensure proper orientation/alignment of the close coil sleeve and spacers.

Similar Events

No previous similar events.

FAILED COMPONENT IDENTIFICATION:

Manufacturer:	General Electric
Manufacturer Model Number:	AMH-4.76-250-1D
Manufacturer Code:	G080
Component Code:	BKR
FitzPatrick Component ID:	71-10504

Corrective Actions**Completed Actions**

Replaced 71-10504 Tie Breaker.

Revised procedures MP-054.01, 4.16 KV Magne-Blast Breaker and MP-054.03, 4.16 KV Magne-Blast Breaker Overhaul, for improved guidance to ensure the proper alignment of the closing coil spacers and brass sleeve or post installation force measurement of the push button actuation to verify the closing assembly operated without binding.

Planned Actions

None.

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1. FACILITY NAME James A. FitzPatrick Nuclear Power Plant		☒ 050 ☐ 052	2. DOCKET NUMBER 05000333		<table border="1" style="width: 100%; border-collapse: collapse;"> <tr> <th colspan="3" style="text-align: left; padding: 2px;">3. LER NUMBER</th> </tr> <tr> <td style="width: 33%; text-align: center; padding: 2px;">YEAR</td> <td style="width: 33%; text-align: center; padding: 2px;">SEQUENTIAL NUMBER</td> <td style="width: 33%; text-align: center; padding: 2px;">REV NO.</td> </tr> <tr> <td style="text-align: center; padding: 2px;">2023</td> <td style="text-align: center; padding: 2px;">– 004</td> <td style="text-align: center; padding: 2px;">– 00</td> </tr> </table>		3. LER NUMBER			YEAR	SEQUENTIAL NUMBER	REV NO.	2023	– 004	– 00
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2023	– 004	– 00													

Safety Significance

Nuclear safety – **There were no actual nuclear consequences.** This tie breaker deficiency is limited to the “A” EDG subsystem and to the 30 day period between March 13 and April 13. The ability of the individual EDG units “A” and “C” to start automatically was not impacted. If the appropriate checks are not achieved, the units will not force parallel; however, the first EDG unit to reach 90% of rated voltage would close onto the emergency power bus while preventing the other unit from closing on the bus. Engineered safeguard loads are automatically connected to the emergency bus in a prescribed sequence. In the case, the load expected during a loss of offsite power (LOOP) event would not challenge a single EDG unit but the worst case Design Basis Accident load for the load sequence may result in stalling that unit.

For a low fuel oil condition as reported by LER-23-003 concurrent with this tie breaker deficiency, it was concluded that there was sufficient fuel oil onsite in all four fuel oil storage tanks and that it could be transferred as needed to run all four EDGs for the required 7 day mission time. The calculation for a 7 day fuel supply assumes full load operation amongst other high standards for performance. This requirement, in conjunction with an ability to obtain replacement supplies within 7 days, supports the availability of EDGs required to shut down the reactor and to maintain it in a safe condition for an abnormal operational transient or a postulated Design Bases Accident with loss of power. The potential effect on nuclear safety related to this report is confined to the period March 27 and April 13, only for the “B” subsystem and only for postulated run times greater than 6 days.

Offsite Power supply remained Operable during the affected period.

References

Issue Report – IR 04669409, ST-9BA Failed Level 1 Acceptance Criteria, dated April 12, 2023

Issue Report – IR 04707644, NRC id’d evaluate potential failure to report, dated October 6, 2023

Issue Report – IR 04711358, PI&R Inspection NRC identified late LER, dated October 20, 2023

Issue Report – IR 04715939, Receipt of finding, dated November 7, 2023

JAF LER – LER: 2023-003, Procedure Error in Emergency Diesel Generator Fuel Oil Supply “B” Subsystem, JAFP-23-0032, dated June 23, 2023